Machine Learning and Regulatory Network Modeling Reveal Immune Drivers of Axillary Lymph Node Metastasis in Breast Cancer

Alba Fischer-Carles[1,4], Tábata Sánchez-Alcántara[2], Yalemzerf M. Alemu[1], Noèlia Gallardo-Borràs[2], Esther Sauras-Colón[2], Alessio Fiorin[2,3], Marylène Lejeune[2,3], Ramon Bosch Príncep[2], Laia Adalid-Llansa[2,4], Daniel Mata Cano[2], Carlos López[2,3], Gemma Piella[1], Jérôme Noailly[1]

[1] BCN MedTech, Department of Engineering, UPF, Spain, alba.fischer@upf.edu
[2] Oncological Pathology and Bioinformatics Research Group, HUTVC, IRB CatSud, URV, Spain
[3] Department of Computer Engineering and Mathematics, URV, Spain
[4] Department of Pathology, HUJ23, ICS, Spain

Introduction

Breast cancer (BC) is the leading cause of cancer death in women, with a worse prognosis when metastasis occurs in axillary lymph nodes (ALN^+) [1]. Immune-tumor interactions are critical in metastatic progression, but mechanisms remain poorly understood [1]. To address this gap, we combine patient data analytics and a regulatory network model (RNM) of the tumor microenvironment (TME) to characterize ALN^+.

Methods

Three random forest (RF) models (1, 2, 3) were trained using clinicopathological and immune data from two luminal BC cohorts (n=83; n=344). Data analytics included feature selection, hyperparameter tuning, and SHAP interpretation. A 4th RF model (Model 4) used features sequentially added by SHAP values. In parallel, a high-level RNM was developed through graph projection of knowledge from 250+ publications, including immune cells, cytokines, hormones, stromal components, and tumor cells, all connected by activation or inhibitory interactions. Node activity was computed through Boolean-based ordinary differential equations [2]. Baseline attractors and key regulators were explored through 100 randomly initialized simulations.

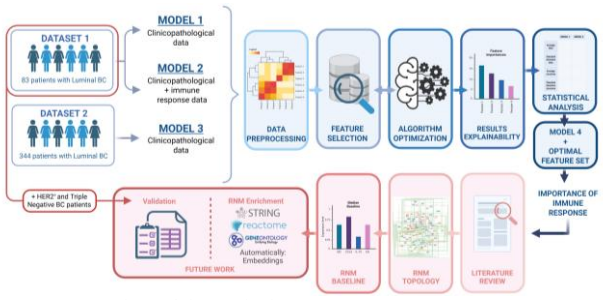


Figure 1: Outline of the study design.

Results

The best model, Model 4 (AUC 0.837; accuracy 0.762; recall 0.716), identified intratumoral $CD21^+$ follicular dendritic cells (DCs) as the strongest predictor of ALN^+ (Fig. 2A, B) [3]. The RNM included 84 nodes and 889 links, with a characteristic path length of 2.189, clustering coefficient of 0.331 and network density of 0.136, reflecting the ability of the RNM to transfer information throughout the different nodes. EMT-tumor-cell node was the most central node, while IFNG was the most central anti-tumoral regulator (Fig. 2C). The baseline represented a stable pro-tumoral phenotype allowing the exploration of anti-tumoral shift strategies (Fig 2D). Notably, IFNG activation induced anti-tumoral immune reprogramming (Fig 2D), consistent with IFNG-driving TME remodeling [4]. The network further reflects the action of macrophages polarization.

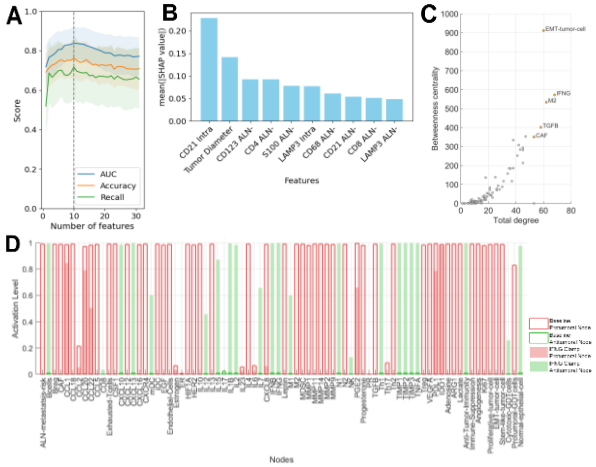


*Figure 2: Results. (**A**) Performance metrics for Model 4. Solid lines: mean values; shaded areas: standard deviations. Dashed line: optimal number of features. (**B**) Mean SHAP values for each feature of Model 4. (**C**) Centrality landscape of regulatory nodes. (**D**) Baseline and IFNG-clamped RNM node activation levels.*

Conclusions

Combining machine learning and biological network modeling identified immune features, particularly intratumoral $CD21^+$ DCs, as key predictors of ALN^+, and highlighted INFG, a potential activator of DCs, as the main driver of anti-tumoral TME reprogramming. Results reveal TME plasticity and therapeutic vulnerability and may guide immune-informed stratification. RNM refinements for an early-stage baseline and validation are sought.

References

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